Advanced Mathematical Techniques (AMT) Notes

1 Today's stuff

Recall the definition of the first Beta function (note the capital):

$$B_1 = \int_0^1 t^{\alpha-1}(1-t)^{\beta-1}dt = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$$

Let's define another Beta function:

$$B_2 : u = \frac{t}{1-t}, u = \frac{1-t}{t} = \frac{1}{t} - 1, t = \frac{u}{u-1}$$

$$B_2 = \int_0^\infty \left[\frac{u}{u+1}\right]^{\alpha-1} \left[\frac{1}{u+1}\right]^{\beta-1} \cdot \frac{t^2}{u^2} du = \int_0^\infty \frac{u^{\alpha-1}}{(u+1)^{\alpha+\beta}}$$

Let's try:

$$\int_0^\infty \frac{u^{\frac{1}{2}}}{(u+1)^3} du = \frac{\Gamma(1.5)\Gamma(1.5)}{\Gamma(3)} = \frac{0.5 \cdot 0.5 \cdot \pi}{2} = \frac{\pi}{8}$$

Also:

$$\int_0^\infty \frac{x^{m-1}dx}{x^n+1}, u = x^n, x = u^{\frac{1}{n}}, dx = \frac{1}{n}u^{1-\frac{1}{n}}du$$

$$\int_0^\infty \frac{u^{\frac{1}{n}^{m-1}}}{u+1}\frac{1}{n}u^{1-\frac{1}{n}}du = \frac{1}{n}\int_0^\infty \frac{u^{\frac{m}{n}-1}du}{u+1} = \frac{1}{n} \cdot \frac{\Gamma(\frac{m}{n})\Gamma(1-\frac{m}{n})}{\Gamma(1)} = \frac{\pi}{n}\csc\frac{m\pi}{n}$$

We're bored, so:

$$\frac{\partial}{\partial m}(\text{above}) = \int_0^\infty \frac{x^{m-1}\ln x}{x^n+1}dx = -\frac{\pi^2}{n^2}\csc\frac{m\pi}{n}\cot\frac{m\pi}{n}$$

Doing it again:

$$\frac{\partial^2}{\partial m^2}(\text{above}) = \int_0^\infty \frac{x^{m-1}\ln^2 x}{x^n+1}dx = \frac{\pi^3}{n^3}\csc\frac{m\pi}{n}\left[2\csc^2\frac{m\pi}{n} - 1\right]$$

Also,

$$\int_0^\infty \frac{x^{\frac{1}{2}}\ln^2 x dx}{x^3+1} = \frac{\pi^3}{27}$$

The above is left as an exercise for the reader.